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# Customer Segmentation Using Clustering

Case Study (Regional Needs): Regional Retail Market Analysis


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# Abstract

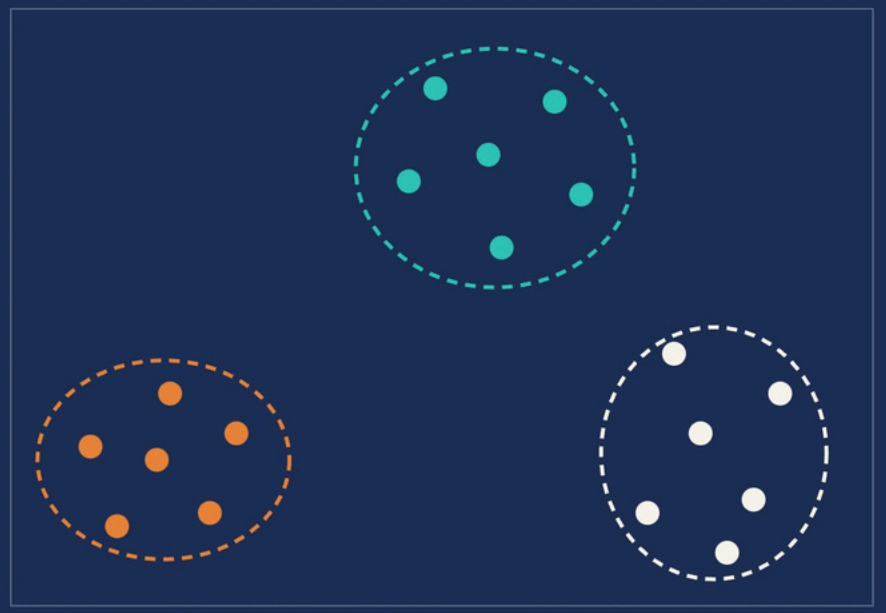
Regional retailers serve customers whose needs differ by town, income and culture, yet most still market to everyone alike. This case study applies unsupervised clustering (K-Means, Hierarchical, DBSCAN) to purchase history, demographics and regional data to discover natural customer groups. Each segment is profiled, validated with Silhouette and Elbow measures, and turned into targeted offers, stocking and pricing decisions that suit regional needs.




**Keywords:** Customer segmentation, K-Means, clustering, regional retail, RFM, market analysis

# Introduction

- **Segmentation** divides customers into groups with similar buying behaviour.
- **Clustering** is unsupervised learning: groups emerge from the data, with no labels needed.
- **Regional need:** preferences vary across urban, semi-urban and rural markets in Tamil Nadu.
- **Inputs:** purchase frequency, spend, product category, location, season.
- **Outcome:** personalised marketing, better stock planning, higher loyalty.



Annual spend vs purchase frequency

Customers grouped into three segments

# Research Gap: Tamil Nadu Survey, 2016 to 2026

Questionnaire and demographic studies

RFM and K-Means in metro and city stores

E-commerce and ML models, few regional ones

## Advantages

- Reveals hidden buying patterns
- Needs no labelled data
- Supports targeted offers

## Disadvantages

- Results vary with the algorithm
- Hard to interpret clusters
- Costly data cleaning

## Limitations

- Most studies cover Chennai, Coimbatore
- Small, single-city samples
- Cluster count chosen manually

## Drawbacks

- Rural and semi-urban needs ignored
- Language and festival effects missed
- Static, one-time analysis

Indicative trends; add your cited papers to support each period.

# Existing System

Regional retailers mostly segment by fixed rules on age, income or location, using manual surveys and spreadsheet reports.

**Manual surveys and sales records** → **Rule-based grouping (age, income)** → **Static Excel reports** → **Same offer for all customers**

## Problems

- Misses hidden behaviour patterns
- Slow, manual and error-prone
- Ignores regional preferences

## Impact

- Poor stock planning and unsold goods
- Low campaign response
- Weak customer loyalty

# Proposed System

A data-driven pipeline that clusters customers by behaviour and region, then validates and refreshes the groups regularly.

1. Collect POS, loyalty, regional data → 2. Clean, scale, RFM features → 3. K-Means, Hierarchical, DBSCAN → 4. Elbow and Silhouette check → 5. Profile segments, dashboard

| Research gap | Proposed solution |
| --- | --- |
| Results vary by algorithm | Compare three algorithms; keep the best score |
| Hard-to-read clusters | Named segment profiles with a simple dashboard |
| Metro-only studies | Cover urban, semi-urban and rural stores |
| Manual cluster count | Elbow and Silhouette pick the number of clusters |
| Static analysis | Periodic re-clustering, with festival and season features |

# Thank you